Predicting Market Salary From LinkedIn Job Posting Data

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Executive Summary

This project develops a machine learning salary prediction tool for full-time job postings in the United States, addressing the need for transparent, data-driven salary estimation. Using a curated dataset of 17,892 full-time positions with salary information, we trained and compared five distinct models:

- Linear Models: OLS, Ridge, Lasso
- Tree-based Models: Decision Tree, Random Forest

Key Findings

- **Non-linear models outperform linear models.**
  Linear regression (OLS, Ridge, Lasso) plateaued at ~36% MAPE, suggesting that salary relationships are not purely additive. The Decision Tree improved performance to 32.8% MAPE, indicating strong nonlinear interactions (e.g., experience × job category × location).
- **Decision Tree provides the best business-aligned accuracy.**
  Although Random Forest achieved slightly lower MAE, the Decision Tree achieved the lowest MAPE (32.8%) and demonstrated the most stable cross-validation performance (±0.6% variance), making it better aligned with relative-error evaluation.
- **Experience level is the strongest salary driver.**
  SHAP analysis shows experience level has over 2.5× the impact of the next most important feature, confirming seniority as the dominant compensation determinant.
- **Job category and technical skills significantly affect salary.**
  Engineering, ML/AI, cloud, finance, and project management skills positively shift salary predictions. Skill extraction from job descriptions adds measurable predictive power beyond job titles alone.
- **Model performance varies by segment.**
  Predictions are most accurate for entry-level roles (MAPE ~29%) and structured categories like Data/ML. Accuracy declines for executive, legal, and healthcare roles due to higher variance and smaller sample sizes.
- **Baseline improvement is substantial but accuracy remains moderate.**
  The final model improves MAPE from 48.1% (median baseline) to 32.8%, a ~15 percentage point improvement. However, predictions remain directional rather than precise, supporting use as a benchmarking tool rather than a definitive pricing mechanism.

1. Problem Definition

1.1 Background and Motivation

As students entering the job market, we frequently rely on LinkedIn job postings to decide where to apply and whether an offer is reasonable. However, salary information on job postings is often missing, inconsistent across similar roles, or difficult to compare across locations and experience levels.

Job market salary transparency is increasingly critical as organizations seek competitive advantage in talent acquisition and retention. However, salary determination remains opaque, driven by subjective factors and market inefficiencies. This creates:

- Information asymmetry: Job seekers lack data-driven benchmarks.
- Inefficient hiring: Recruiters rely on intuition rather than systematic analysis.
- Equity concerns: Potential for unconscious bias in compensation decisions.

1.2 Research Objectives

We propose a proof-of-concept salary benchmarking tool that predicts inflation-adjusted annual salaries using structured job posting features available at prediction time.

This project aims to:

- Predict salary outcomes for full-time job postings using machine learning.
- Understand the key drivers of salary variation through feature importance analysis.
- Quantify prediction uncertainty via confidence intervals.
- Enable HR teams to make data-driven compensation decisions.

1.3 Scope

We focus on full-time job postings in the United States, using postings collected during 2023 to 2024 as the primary data window. We define full-time using the posting work type field and keep only records labeled as full-time. We define the prediction target as an annualized salary so that compensation levels are comparable across postings. To make estimates more relevant to the current market, we apply a simple inflation adjustment using an assumed 3.5% annual rate compounded over two years (about a 7.1% total adjustment) to approximate 2025 to 2026 levels. Because the source is LinkedIn job postings, results should be interpreted as LinkedIn-market estimates rather than representative labor-market wages.

1.4 Impact

Poor salary benchmarks can lead to mispriced offers, longer negotiation cycles, and lower trust between candidates and employers. When salary expectations rely on incomplete or inconsistent posting information, candidates may apply to fewer suitable roles or negotiate from inaccurate anchors, while employers may lose strong applicants due to uncompetitive offers and inefficient compensation decisions. From a modelling perspective, this is also a challenging prediction task. The salary target is strongly right-skewed, the distribution varies widely across work type, state, and experience level, and the limited share of labeled postings increases the risk of selection bias. As a result, simple heuristics are unlikely to generalize well across segments, which motivates a structured machine learning approach supported by careful data cleaning and feature construction.

1.5 Intended uses and use case

The tool supports three common decision settings. Job seekers use the predicted salary band to prioritize applications and form a negotiation anchor. Recruiters and hiring managers use it to benchmark roles when postings do not include pay ranges, which can shorten negotiation and improve offer competitiveness. Compensation teams use it to compare roles across locations and experience levels for planning and internal benchmarking, especially when historical pay ranges are limited.

1.6 Modelling Task

We formulate the task as a supervised learning regression problem. The input includes job title and job description text, state as the location, work type as full-time, experience level, and whether the role is remote or on-site. The target is the inflation-adjusted annualized salary. Given these inputs, the model produces a point estimate of annual salary and translates held-out test performance into an uncertainty range around that estimate. In practice, we use the distribution of test-set residuals to construct prediction intervals, such as 80% and 90% intervals, and we report the corresponding salary range. We also evaluate how often these intervals contain the true salary on the test set, so the reported range and its empirical coverage together serve as an interpretable confidence signal for decision support.

1.7 Evaluation Metrics and Success Threshold

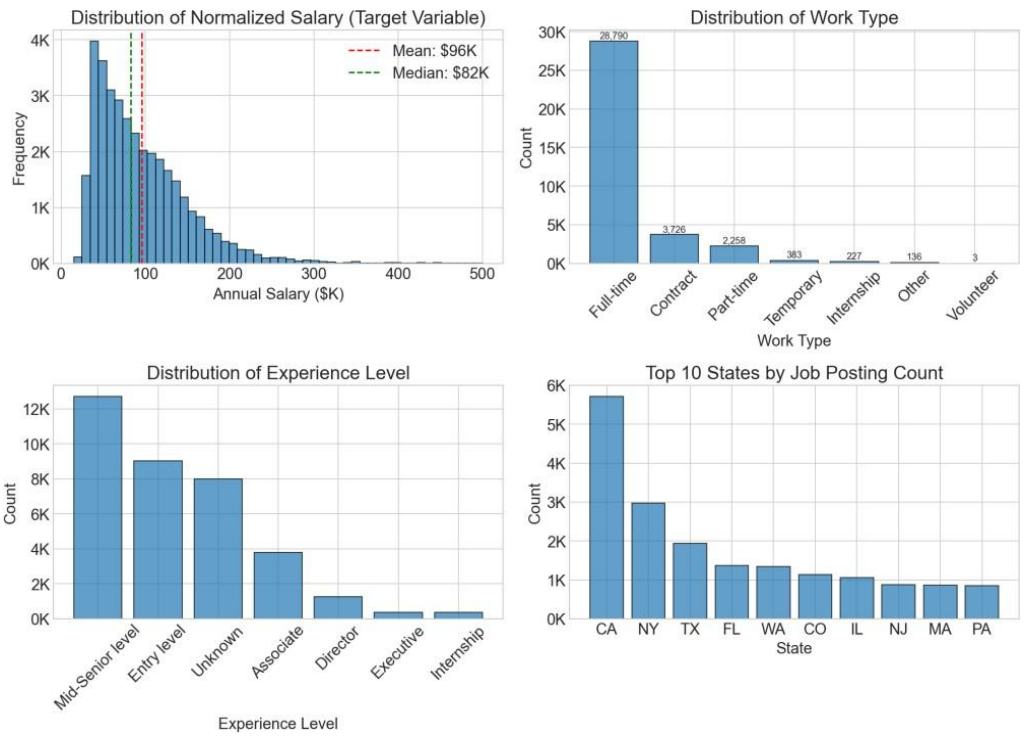
To align evaluation with the business use case, we use error metrics that communicate how far predictions deviate from true salaries in relative terms. MAPE is the primary metric because it is easy to interpret and aligns with communicating practical salary ranges. We report MAE and RMSE as complementary measures, while R-squared is included for reference. We benchmark all models against a simple baseline predictor. In an initial pre-cleaning run, a mean baseline performed poorly with about 60% test-set MAPE. After applying the final filtering and cleaning pipeline, the median baseline improves to 48.1% test-set MAPE. Our minimum success criterion is a clear improvement over the final baseline, and our stretch goal is to reach test-set MAPE around the low-30% range on the final cleaned dataset. Based on the initial data review, we restrict the modeling dataset to full-time postings with valid salary and location fields, then apply a consistent cleaning and feature pipeline before comparing model families.

2. Dataset

Our analysis uses the LinkedIn Job Postings dataset available on Kaggle (Arsh Kon, "LinkedIn Job Postings"), which contains structured job listing data scraped from LinkedIn. The dataset provides rich structured and text-based information suitable for supervised salary prediction. Key characteristics include:

- **Size and scope:** Over 120,000 job postings across multiple U.S. states and industries.
- **Salary fields:** Minimum, median, and maximum salary; pay period (hourly/yearly); currency; compensation type; and a normalized annual salary variable.
- **Job attributes:** Job title, company name, location, work type (e.g., full-time, internship), experience level, remote eligibility.
- **Text data:** Detailed job descriptions and skills descriptions, enabling skill extraction and feature engineering.
- **Engagement metrics:** Views, number of applications, posting timestamps, and listing status.
- **Geographic indicators:** ZIP code and FIPS codes for regional analysis.

Figure 1: Raw Data Overview (Pre-cleaning)



3. Methodology

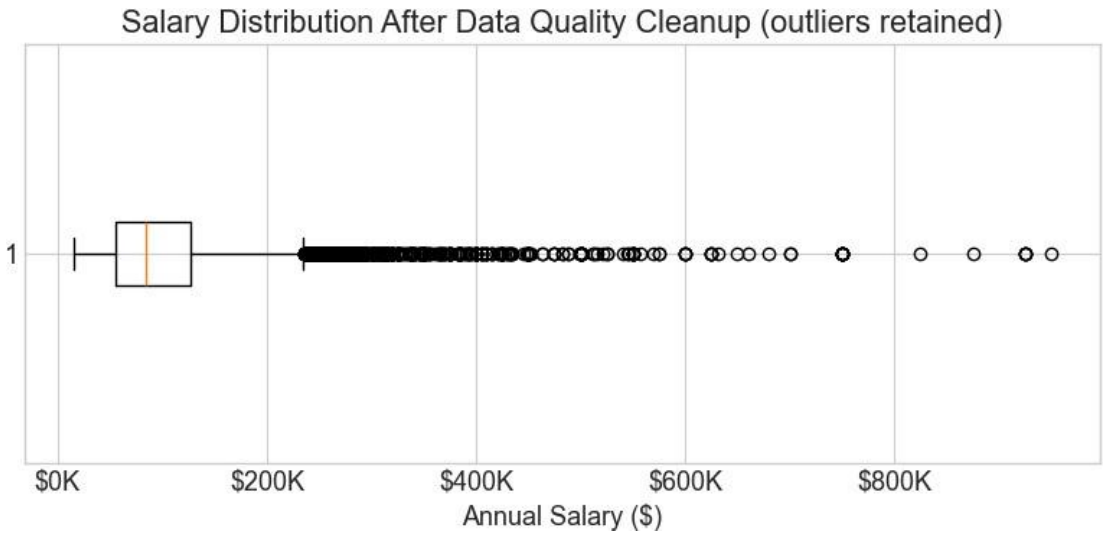
Based on the preliminary analysis results, we decided to narrow the scope of predictions to full-time job postings with a listed salary. Below are the procedures we took to explore the best possible pipeline.

3.1 Feature Engineering

1. Data cleaning

- **Row selection**
  First, we kept only job postings that are full-time, list a non-zero salary, and have location information, reducing the dataset from 123,849 to 23,854.
- **Salary data fixing**
  Then we fixed mislabeled pay periods, such as changing a $22 yearly rate to a $22 hourly rate to correct the annualized salary. After a sanity check of the outliers, we also removed postings with annual salaries below $15K (the minimum wage) or above $1M (without entry-level titles). To account for the nuances of predicting the 2025-2026 salary range using the 2023-2024 dataset, we also scaled the salary data using an annualized 3.5% CPI compounding over 2 years (factor = 1.07) to normalize the existing salary data to 2025-2026 levels (Mercer, 2024).

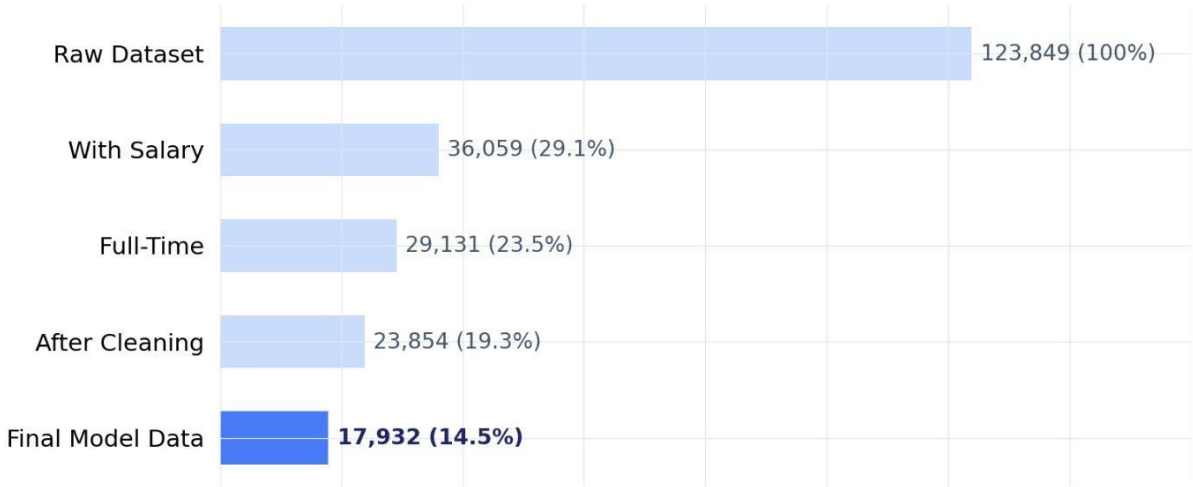
Figure 2: Salary Distribution After Data Quality Cleanup



- **Missing value handling**
  Thirdly, we filled in the missing experience label using keyword extraction from job titles, and the remote label using keyword extraction from job descriptions, assuming no remote work is allowed unless stated otherwise.

- **Adding company information**
  Lastly, we joined the posting dataset with the other 3 datasets on company industry and size to include metrics about the company, such as its industry, number of employees, and LinkedIn follower counts.

Figure 3: Data Cleaning Funnel



2. One-hot encoding existing categorical features

For the existing categorical features in the dataset, experience level, state and industry, we conducted one-hot encoding.

3. Create new features

Intuitively, the type of job and skills required would significantly influence salary range, but this metric is missing from the job posting dataset. We created 19 job categories based on keyword matches in job titles and 16 skills derived from keyword extraction in job descriptions. See the table below for more information

Table 1: *Grouping Jobs into 1S Job Categories*

#	Job Category	Count	%	Sample Job Titles (matched by keywords)
1	Engineering/Trades	3,776	15.8%	Mechanical Engineer, HVAC Technician
2	Other	3,460	14.5%	*(Titles not matching any keyword pattern)*
3	Management/Director	2,484	10.4%	General Manager, Operations Director
4	Healthcare/Clinical/Nursing	2,267	9.5%	Registered Nurse (RN), Physician
5	Administrative	2,096	8.8%	Administrative Assistant, Office Coordinator
6	Finance/Accounting	2,067	8.7%	Financial Analyst, Staff Accountant
7	Sales	1,585	6.6%	Account Executive, Sales Representative
8	Data/ML/Analyst	1,334	5.6%	Data Scientist, Data Engineer, ML Engineer
9	Software/Dev/IT/DevOps	969	4.1%	Software Engineer, Full Stack Developer
10	Retail/Customer Service	620	2.6%	Customer Service Rep, Retail Store Associate
11	Logistics/Warehouse	590	2.5%	Warehouse Associate, Delivery Driver
12	Project Management	534	2.2%	Project Manager, Program Manager

#	Job Category	Count	%	Sample Job Titles (matched by keywords)
13	Marketing	410	1.7%	Marketing Manager, SEO Specialist
14	Legal	404	1.7%	Attorney, Paralegal, Compliance Officer
15	HR/Recruiting	394	1.7%	HR Manager, Talent Acquisition Specialist
16	Product/Design/UX	385	1.6%	Product Manager, UX Designer
17	Hospitality/Food	204	0.9%	Chef, Restaurant Manager, Housekeeper
18	Education	182	0.8%	Teacher, Professor, Instructor, Tutor
19	Consulting	116	0.5%	Management Consultant, IT Consultant

Table 2: *Extracting 1c Skills from Job Description*

#	Skill Feature	Count	%	Keywords Matched
1	skill_management	15,711	65.8%	management, managing, leadership, supervising
2	skill_communication	12,941	54.2%	communication, presentation, interpersonal
3	skill_healthcare	11,843	49.6%	nursing, patient care, clinical, medical, HIPAA
4	skill_finance	8,787	36.8%	finance, accounting, budget, financial

5	skill_sales	5,611	23.5%	sales, business development, client relations
6	skill_marketing	5,205	21.8%	marketing, SEO, content, social media, brand
7	skill_engineering	5,205	21.8%	engineering, CAD, AutoCAD, mechanical
8	skill_excel	4,250	17.8%	excel, spreadsheet
9	skill_project_mgmt	2,985	12.5%	project management, PMP, agile, scrum
10	skill_data_analysis	2,077	8.7%	data analysis, analytics, tableau, power bi
11	skill_cloud	1,576	6.6%	aws, azure, gcp, cloud
12	skill_ml_ai	931	3.9%	machine learning, deep learning, AI, ML
13	skill_sql	812	3.4%	sql
14	skill_python	812	3.4%	python
15	skill_javascript	478	2.0%	javascript, js, react, angular, node.js
16	skill_java	406	1.7%	java

3.2 Performance Metric Selection

We decided to use MAPE on test set <= 30% as the key metric for model success the following reasons.

Interpretability for stakeholders

MAPE expresses error as a percentage relative to actual salary, such as "the model gives you a salary prediction 70%~130% of the actual salary", which is immediately meaningful to job seekers and HR professionals. RMSE or MAE, on the other hand, is less intuitive for stakeholders.

Penalizes errors across the segment equally

MAPE penalizes errors in high- and low-salary ranges equally. For example, a 30% prediction error means the same for both $50K dollar job and $100K dollar job, whereas the RMSE of $30,000 is almost 60% off for the $50K job but only accounts for 30% off for

the $100K job. Using MAPE as the key metric would prevent the model from sacrificing accuracy on lower-paid roles to optimize for high-salary predictions.

Model Class Selection

We used 6 model classes, ranging from simple to complex.

1. Baseline: predict the median

Based on the exploratory analysis shown in the previous section (dataset), we observed that the salary distribution is highly right-skewed, indicating that the mean is significantly higher than the median. So we used the median prediction as a baseline.

2. Linear Model Class

The first model class we tried is the linear model, assuming that features captured drive salary additively and independently, with more interpretable coefficients for each feature. For example, if linear models give us the result with a +$5,000 salary associated with a higher experience level, and having MAPE = 30%, it is more intuitive to market towards stakeholders with a key message such as “our tool tells you how much more you can earn with an advanced title, the result is only off by 30% at most”.

OLS Linear Regression

We started with the simplest linear regression model without regularization, which gave us an MAPE of 35.9% and a 12.2 ppt improvement over baseline, but it did not meet the 30% threshold.

Ridge Regression

We then tried Ridge regression with regularization, which handles multicollinearity from 100+ one-hot encoded features by shrinking all coefficients to zero. This gives us an MAPE of 36%, only a very slight improvement over OLS linear regression, and it still does not meet the 30% threshold.

Lasso Regression

We also tried Lasso regression with regularization, which performed feature selection by setting irrelevant coefficients to zero. Lasso retained almost all features, zeroing out only 32 of 228, indicating that all features are important for salary prediction. As a result, it gave us almost the same MAPE(35.9%) as Ridge regression.

3. Tree Model Class

After trying multiple linear models, we did not see a significant improvement in MAPE, indicating that the effect of features on salary prediction is probably nonlinear. We also tried three models to improve accuracy, but at the cost of less intuitive, quantifiable interpretations for stakeholders.

Decision Tree

The decision tree model yields an MAPE of 32.8%, significantly improving on the baseline and the linear model class as well. Indicating that the tree model captures nonlinear and interactive relationships among features that affect salary prediction. The result matched stakeholders' intuition: for example, software engineers in California might earn much more than those in Texas, but the same might not hold for nurses. But the decision tree model has not met the 30% of threshold either.

Random Forest

So the last model we tried is a random forest to investigate whether our decision tree model has high variance (overfitting). The random forest model has an MAPE of 34%, slightly higher than the decision tree model, and performs better than linear models. This result indicates that the decision tree model is not overfitting

Below is a table summarizing the advantages and limitations of the 5 non-baseline models we tried:

Table 3: Advantages and Limitations of Each Model for Salary Prediction.

Model Class	**Model**	**Advantages**	**Limitations**
Linear	**OLS**	Shows exact dollar impact of each feature (e.g., "Director adds $X to salary"); p-values identify statistically significant predictors	Cannot capture that skills affect salary differently across job categories
	Ridge	Stabilizes coefficients across 100+ one-hot encoded state/category/industry dummies	Still assumes each feature adds a fixed dollar amount regardless of context
	Lasso	Identifies which of 100+ encoded features actually matter for salary; zeros out irrelevant states/industries	May arbitrarily drop one correlated feature (e.g., keeps CA but drops NY

			despite both being significant)
Tree	**Decision Tree**	Learns rules like "Software Engineers with cloud skills in CA earn $180K+"; SHAP explains individual salary predictions	May memorize training salary patterns; sensitive to which job postings are in the training set
	Random Forest	Most accurate salary predictions by averaging 250 diverse trees; robust feature importance across job categories	Harder to explain *why* a specific salary was predicted; cannot predict salaries beyond the $15K–$1M training range

Model Hyperparameter Tuning

All 5 non-baseline models are tuned using 5-fold cross-validation on the training dataset; the test set is held out entirely and only used for final evaluation.

The table below summarizes how we split the dataset for each model:

Table 4: Advantages and Limitations of Each Model for Salary Prediction.

Model	Train	Validation	Test
OLS	Fit on full 80% training set	No validation (no hyperparameters to tune)	20% hold-out
Ridge	In each CV fold: fit on 4/5 of training set (64% of total)	1/5 of training set (16% of total), rotated across 5 folds to select best alpha	20% hold-out
Lasso	Same as Ridge	Same as Ridge, selects best alpha	20% hold-out
Decision Tree	In each CV fold: fit on 4/5 of training set (64% of total)	1/5 of training set (16% of total), rotated across 5 folds to select best hyperparameter combination	20% hold-out

Random Forest	Same as Decision Tree	Same as Decision Tree	20% hold-out

For Ridge and Lasso regression, we used scikit-learn's built-in CV classes, which efficiently evaluate all alpha candidates in a single call. Features were standardized before fitting, since Ridge and Lasso penalties are scale-sensitive.

For tree models, we used GridSearchCV to search all parameter combinations with 5-fold CV. The Decision Tree grid explored 900 combinations (6×5×5×2×3) and the Random Forest grid explored 1,200 combinations (5×6×5×4×2), each multiplied by 5 folds. All grid searches ran in parallel to reduce computation time.

As a result, below is the best hyperparameter we obtained for each model:

Table 5: Hyperparameter obtained for each model

Model	Best Hyperparameters	Tuning Method
OLS	None (no hyperparameters)	N/A
Ridge	$\alpha = 1{,}151$	RidgeCV, 5-fold CV over 50 alphas (10^{-2} to 10^{6})
Lasso	$\alpha = 115$; features retained: 196/228 (86%)	LassoCV, 5-fold CV over 50 alphas (10^{-1} to 10^{5})
Decision Tree	max_depth=15, min_samples_split=50, min_samples_leaf=4, criterion=absolute_error, max_features=None	GridSearchCV, 5-fold CV, 900 combinations
Random Forest	n_estimators=250, max_depth=None, min_samples_split=2, min_samples_leaf=1, max_features=sqrt	GridSearchCV, 5-fold CV, 1,200 combinations

4. Result

In this section, we will discuss how we chose the decision tree model as the best performer.

4.1 Model Performance

As previously discussed, we chose the MAPE on the test set as the core metric to better address stakeholders' concerns about "How far off is the prediction, relative to the salary itself?"

Below is a chart summarizing the performance of all 6 models on the test set, with MAPE as the key metric and MAE/RMSE/R^2 as the reference metric.

Figure 4: Model Performance Comparison

All models outperform the baseline (baseline MAPE = 48.1%). Every model reduced MAPE by 12% to 15%, confirming that the features captured are meaningful for predicting salary ranges and that an advanced prediction method is necessary.

Linear models perform nearly identically (MAPE 35.G–36.0%). OLS, Ridge, and Lasso produce similar MAPE results, indicating that multicollinearity is not a major issue, and most of the 228 features contribute some signal (Lasso zeroed only 32 of 228). Indicating that the problem is not overfitting in the linear class, but it's the linear assumption itself.

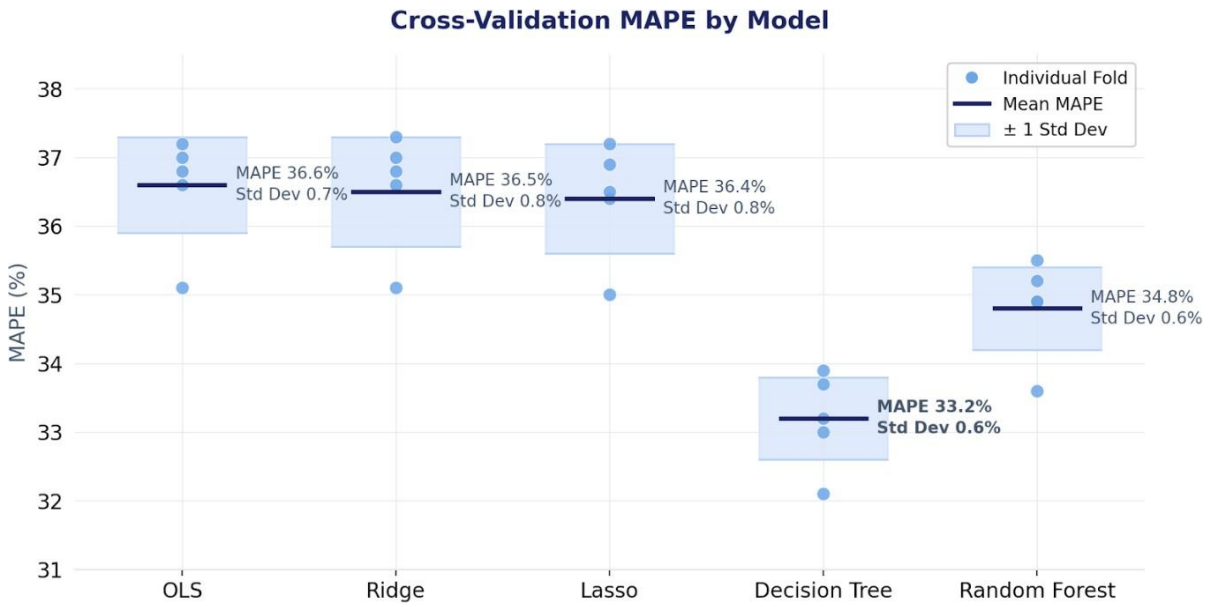
The Decision Tree achieves the best MAPE (32.8%) but also the highest RMSE ($50,640). This is a 15.3 pp improvement over the baseline, meaning the model's predictions are on average within ~33% of the actual salary. This suggests salary is driven by non-linear interactions (e.g., experience level × job category) that linear models cannot capture. This indicates that Decision Tree occasionally makes large dollar errors on high-salary outliers, but its percentage-based accuracy across all salary ranges is superior.

Random Forest has the lowest MAE ($30,142) but not the lowest MAPE (34.0%). This means that Random Forest makes smaller absolute errors overall, but Decision Tree is more accurate proportionally for lower-salary jobs. Since MAPE weights errors relative to salary, the Decision Tree better serves the use case of predicting salaries across all salary levels fairly.

4.2 Model Robustness

Another question we think our stakeholders would be interested in is: "How stable is the prediction, relative to job posting updates?" We also conducted a 5-fold cross-validation to determine whether the decision tree remains the most robust, and the results are summarized below.

Figure 5: Model Robustness Comparison



Decision Tree's five shows the tightest clustering (lowest variance). They span only ~1.8 percentage points (32.1%–33.9%), compared with ~2.2 percentage points for OLS/Ridge/Lasso (35.1%–37.3%) and ~1.9 percentage points for Random Forest (33.6%–35.5%). The tight clustering, with a standard deviation of ±0.6%, indicates that the Decision Tree generalizes consistently across different subsets of the data and is not overfitting to any particular fold. This stability is essential for salary prediction when job postings are updated over time.

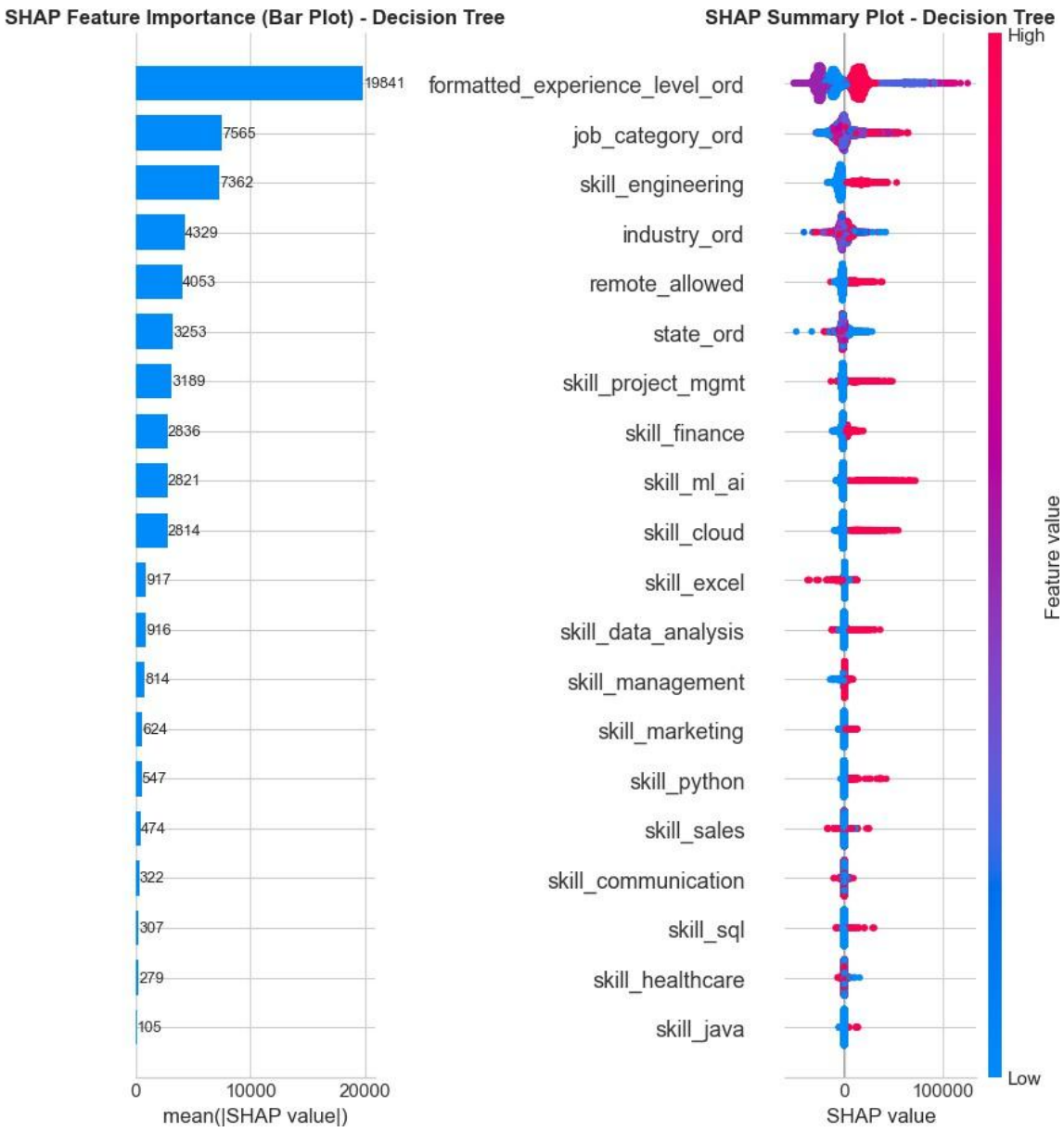
4.3 Model Interpretation

After we have built the decision model for our salary predictor, we will interpret the results in this section.

Feature Importance: What is driving salary prediction?

To understand the importance of each feature for salary prediction, we plotted the SHAP values shown below.

Figure c: SHAP Plot of the Decision Tree Model



- To quantify how each feature contributes to the model's salary predictions, we computed SHAP values for the Decision Tree model. The bar plot (left) shows the mean absolute SHAP values, which measure each feature's overall importance regardless of direction. The summary plot (right) reveals the directionality, how high or low feature values push predictions up or down.

- *Experience level* dominates all other features with a mean absolute SHAP value of 19,841, more than 2.5 times the next most important feature. This confirms that seniority is the single strongest determinant of predicted salary, consistent with labour-market expectations that compensation scales sharply with career progression.

- *Job category* (7,565) and *engineering skill* (7,362) follow as the second- and third-most important features. Notably, the engineering skill metric, a binary feature we extracted from job descriptions, carries nearly as much predictive weight as the entire job category variable. This validates our feature engineering approach: keyword-based skill extraction from unstructured text captures a meaningful salary signal that a job title alone misses.

- *Industry* (4,329) and *remote* work allowance (4,053) form a second tier of importance. The summary plot shows that remote-allowed positions (high feature value, shown in red) have higher salary predictions, which may be because most of them are in tech industries.

- Among individual skill features, *project management* (3,189), *finance* (2,836), *ML/AI* (2,821), and *cloud* (2,814) are the most impactful. The summary plot shows that these skills exhibit a clear positive directionality: having these skills increases predicted salary, while their absence has a near-zero or slightly negative effect. This asymmetry suggests these are "nice-to-have skills" that contributed to a higher salary rather than baseline requirements.

- The long tail of remaining skill features (Excel, data analysis, management, marketing, Python, SQL, etc.) each contributes modestly, but collectively adds meaningful predictive power. Their inclusion reduces overall MAPE compared to a model trained without skill features, confirming that granular skill extraction from job descriptions is a worthwhile addition to the feature set despite individual features having relatively small SHAP values.

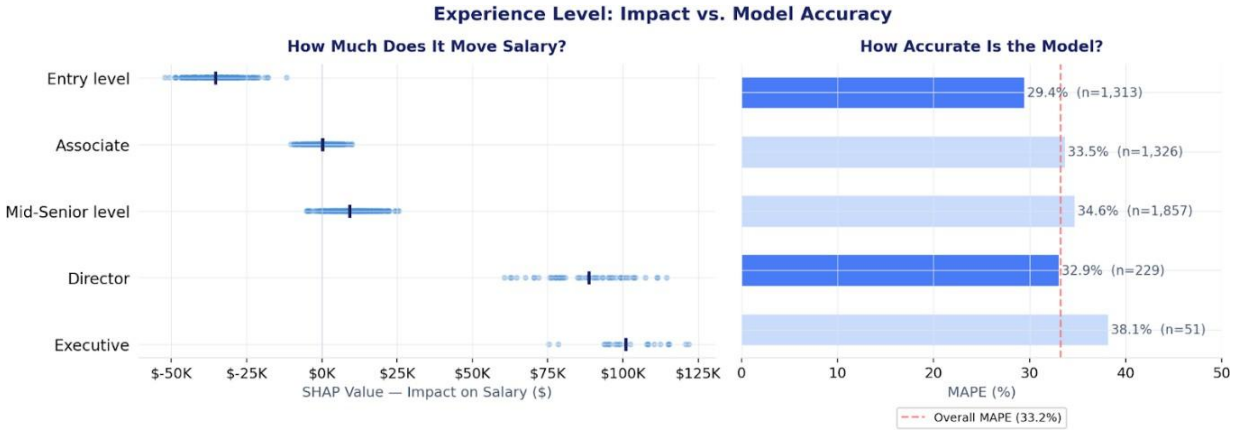
Use Cases: Who is more suitable to use this predictor?

To understand not only what the model predicts but also where predictions are most and least reliable, we conducted SHAP analysis and computed MAPE for the two most important features, experience level and job category.

The current model is more accurate in predicting entry-level job salaries.

Experience Level is the single most influential feature, accounting for approximately 40% of the model's decision-making. From the figure below, we see that SHAP analysis tells us Executive and Director roles raise predicted salaries by $75,000–$100,000, while Entry-level roles lower them by $25,000–$50,000. However, the model is the most accurate for entry-level salary predictions (MAPE 29.4%, n=1,313), likely because it benefits from a large sample and relatively standardized compensation bands. Executive predictions are the least reliable (MAPE 38.1%, n=51), as the small sample size and high salary variance, for example, executives at a startup versus a Fortune 500 company may differ by hundreds of thousands of dollars, limit the model's ability to generalize across different experience levels.

Figure 7: Model Result by Job Posting's Experience Level

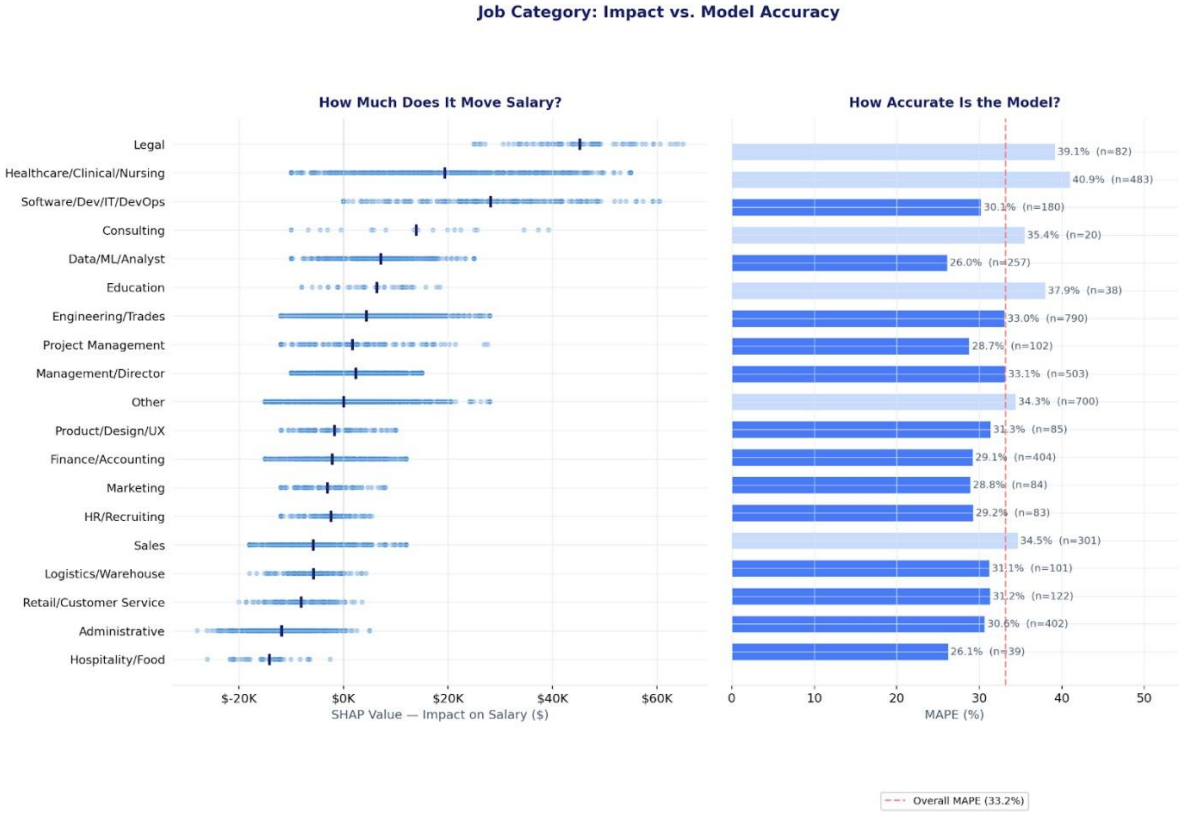


The current model is more accurate in predicting well-defined job categories.

Job Category is the second most important feature, and the SHAP analysis shows a wide range of salary impacts. Legal, Healthcare/Clinical/Nursing, and Software/Dev/IT/DevOps roles are the strongest positive salary drivers, with SHAP values ranging from $40,000 to $65,000 above baseline. At the other end, Administrative and Hospitality/Food roles push predictions downward by $10,000–$20,000. Categories where the model performs best, Data/ML/Analyst (26.0%) and Hospitality/Food (26.1%), tend to have well-defined, narrow salary bands within their category. Categories where the model is less accurate, such as Healthcare/Clinical/Nursing (40.9%) and Legal (39.1%), indicate that the current model does not effectively capture features that affect their salary range. This could be because jobs within these categories vary widely; for example, a "Healthcare" posting could range from a medical

receptionist to a nurse practitioner, and a "Legal" posting could range from a paralegal to a corporate attorney.

Figure 8: Model Result by Job Posting's Job Category



5. Practical Implication

5.1 Business Value

The salary prediction framework transforms scattered job posting data into structured compensation benchmarks that support data-driven decision-making.

- **HR and Compensation Teams:** Use predicted salary ranges to benchmark roles, design compensation bands, assess internal pay equity, and estimate hiring budgets under uncertainty.
- **Recruiters and Hiring Managers:** Validate whether offered salaries align with market rates, reduce reliance on intuition during negotiations, and improve offer competitiveness.
- **Job Seekers:** Gain greater salary transparency to assess market value, benchmark offers, and negotiate compensation based on skills, experience, and location.

5.2 Use Cases and Applications

- **Salary Range Generator:**
  Input job attributes (role, experience level, location, skills) → output predicted salary range with confidence interval.
- **Internal Pay Equity Auditing:**
  Compare employee salaries against predicted market benchmarks to identify systematic over- or under-compensation.
- **Market Monitoring:**
  Aggregate predicted salaries by job category, skill, or geography to track compensation trends over time.
- **Hiring Scenario Simulation:**
  Estimate projected hiring costs across multiple roles to support budgeting and workforce expansion planning.

5.3 Model Limitations

Despite its usefulness, the model has important limitations:

- **Selection Bias:** Only job postings with disclosed salaries are included, which may overrepresent certain industries or regions.
- **Moderate Accuracy:** With ~30% MAPE, predictions should be interpreted as directional estimates rather than precise salary amounts.

- **Limited Generalization:** Performance declines for extreme salaries and highly specialized executive roles due to limited data.
- **Feature Sensitivity:** Missing or incomplete job descriptions can affect prediction accuracy.
- **Dynamic Labor Markets:** New roles and emerging skills may not be reflected in the training data, requiring periodic retraining.
- **Ethical Considerations:** Since the model is trained on historical salary data, it may reflect existing structural inequities. Outputs should assist—not replace—human judgment.

5.4 Recommendations and Next Steps

- Deploy the model as a decision-support tool, presenting salary ranges with explicit uncertainty indicators rather than single-point estimates.
- Establish a feedback loop to compare predicted salaries with actual accepted offers, enabling continuous retraining and performance monitoring.
- Maintain transparency by clearly communicating model uncertainty and known error patterns.
- Expand the model to include broader compensation components (bonuses, equity) for more comprehensive predictions.
- Explore category-specific models to improve accuracy within major job families.

6. References

Mercer. (2024). *Uncover Mercer's U.S. compensation planning survey results.* https://www.mercer.com/en-us/insights/total-rewards/compensation/us-compensation-planning-survey/